Masoud Nateghi

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EDUCATION

- **Ph.D. in Computer Science** *Aug 2023 - May 2028*
  Emory University, Atlanta, GA
  GPA: 4.00/4.00
- **B.Sc. in Electrical Engineering** *Sep 2017 - Sep 2022*
  Sharif University of Technology, Tehran
  GPA: 17.18/20.00

RESEARCH EXPERIENCE

The Alphanumerics Lab, Emory University *Aug 2023 – Present*
PhD Researcher *Advisor: Dr. Reza Sameni*

- **Physiological Biomarker Discovery for Major Depressive Disorder (MDD)**
  — Head of Biomedical Data Collection Team, Emory University
  - Led multimodal biomedical data collection team at Emory site, developing guidelines for high-quality data acquisition of EEG, ECG, PPG, eye-tracking, electrodermal activity, respiration, and continuous BP recordings from over 30 participants in a psycholinguistic experiment funded by DARPA called *PRESCRIBE*.
  - Processed wearable (ECG/PPG) and neural (EEG) signals, extracted ERP-derived neural features (LFP, N400, LPP) and cardiac biomarkers, and developed ML pipelines to predict depression and its severity across depressive and healthy individuals.
  - Research efforts resulted in five manuscripts (two published, one preprint, and two under review).
- **Fetal ECG for Prenatal Health Monitoring**
  - Curated the first and largest noninvasive fetal ECG dataset with clinically annotated pregnancy outcomes using the FDA-cleared MERIDIAN fetal monitor from multiple international sites.
  - Implemented a generic Kalman filter–based pipeline for causal inference to detect instantaneous maternal-fetal couplings with open-access source code.
  - Developing a foundation model tailored to fetal ECG capable of downstream tasks including fetal ECG extraction, fetal R-peak detection, and latent representation–based phenotyping of adverse pregnancy outcomes using WaveNet, U-Net, and transformer-based architectures.
  - Awarded the CAIRE Fellowship at Emory University to support the development of FECG foundation models.
- **Cardiovascular Signal Analysis for Chronic Kidney Disease (CKD)**
  - Developed an end-to-end Python/MATLAB pipeline for automated processing and analysis of ECG, continuous blood pressure (BP), and muscle sympathetic nerve activity (MSNA) recordings in CKD patients and healthy controls.
  - Implemented signal processing algorithms for fiducial point detection, MSNA burst detection, signal quality assessment, and extraction of cardiovascular and autonomic biomarkers, including heart rate variability, cardiac baroreflex sensitivity, and sympathetic nerve activity.
  - Built an interactive web-based platform integrating data management, batch processing, signal visualization, and biomarker extraction, to enable clinicians with limited programming experience to perform automated signal analysis without writing code.

PUBLICATIONS

Journal Papers

1. **Masoud Nateghi**, Reza Sameni. *An Estimation Framework for Cuff-based Blood Pressure Measurement: Analysis and Correction of Systematic Biases.* Physiological Measurement (2026). DOI: 10.1088/1361-6579/aebde2

2. **Masoud Nateghi**, Reza Sameni. *A Kalman Filter–Based Framework for Granger Causality Assessment: Application in Tracking Maternal–Fetal Heart Rate Coupling.* IEEE Transactions on Biomedical Engineering (2026). DOI: 10.1109/TBME.2026.3710372
3. Sajjad Karimi, **Masoud Nateghi**, et al. *Prescreening Depression Using Wearable Electrocardiogram and Photoplethysmogram Data from a Psycholinguistic Experiment.* Physiological Measurement (2025). DOI: 10.1088/1361-6579/adf6fe
4. Bradly T. Stone, Phillip C. Desrochers, **Masoud Nateghi**, et al. *Decoding Depression: Event-Related Potential Dynamics and Predictive Neural Signatures of Depression Severity.* Journal of Affective Disorders (2025). DOI: 10.1016/j.jad.2025.119893
5. **Masoud Nateghi**, Mahdi Rahbar Alam, Hossein Amiri, Samaneh Nasiri, Reza Sameni. *Model-Based Electroencephalogram Instantaneous Frequency Tracking: Application in Automated Sleep–Wake Stage Classification.* Sensors (2024). DOI: 10.3390/s24247881

Conference Papers

1. Johann Vargas-Calixto, **Masoud Nateghi**, et al. *Self-Supervised Representation Learning of Fetal Heart Rate Outperforms a Foundation Model in Identifying Fetal Acidosis.* Computing in Cardiology (2026).
2. **Masoud Nateghi**, Reza Sameni. *Estimation-Theoretic Bias Reduction for Oscillometric Blood Pressure Readings.* Proceedings of the AMIA Annual Symposium (Accepted, 2026). Preprint DOI: 10.48550/arXiv.2508.15687
3. Seibi Kobara, Alireza Rafiei, **Masoud Nateghi**, et al. *Social Media as a Sensor: Analyzing Twitter Data for Breast Cancer Medication Effects Using Natural Language Processing.* Artificial Intelligence in Medicine (AIME) (2024). DOI: 10.1007/978-3-031-66538-7_34

Preprints

1. Reza Sameni, Gabriela I. Cestero, **Masoud Nateghi**, et al. *A Psycholinguistics Protocol with Simultaneous Multimodal Physiological Data Collection for Individualized Pre-Screening Depressive Disorders.* protocols.io. DOI: 10.17504/protocols.io.dm6gpme3dgzp/v1

TECHNICAL SKILLS

Programming:	Python, MATLAB, C/C++, Java, Verilog, Assembly
Computing & HPC:	Linux, Slurm, Bash, SSH, HPC Clusters
Development Tools:	GitHub, Copilot, Codex, Calude, ChatGPT
Scientific Libraries:	PyTorch, TensorFlow, Scikit-learn, MNE, EEGLAB, NeuroKit, OSET
Core Expertise:	Digital Signal Processing, ML/DL, Kalman Filtering, Estimation Theory

HONORS AND SERVICES

- Recipient of CAIRE Fellowship at Emory University
- Ranked 167th among 150,000 participants in Iran's National University Entrance Exam
- Journal reviewer: *Physiological Measurement*, *Biomedical Signal Processing and Control*, *Discover Medicine*